RIYAD W. HASSEN

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RESEARCH INTERESTS

Multimodal reasoning in **vision-language models**; **multilingual and low-resource** language technologies, with a focus on African languages; **retrieval-augmented generation**; efficient and **hybrid model architectures** (Transformer/state-space).

EDUCATION

MSc (Computer Sciences) University of Wisconsin–Madison, WI, USA *Aug 2023 – May 2026*
Coursework: Foundation Models, Deep Learning, NLP, Computer Vision, Machine Learning, Agentic AI, Artificial Intelligence, Big Data

BSc (Software Engineering) Addis Ababa University, Ethiopia *Sep 2016 – Oct 2021*
Graduated with Distinction. Thesis: Crop yield prediction with machine learning and deep learning (published; see below)

PUBLICATIONS

- **R. W. Hassen**. Crop yield prediction using machine learning and deep learning. *Addis Ababa University Journal*, 2021. (Undergraduate thesis) PDF

RESEARCH EXPERIENCE

UW–Madison Graduate Researcher — Madison, WI *Jan 2025 – May 2025*
Detecting Degenerative Cervical Myelopathy (DCM) from video pose data

- Designed a **Transformer-based time-series** architecture to detect **DCM**, a spinal cord disorder, from pose sequences extracted from patient video.
- Built the **end-to-end ML pipeline**: pose-sequence preprocessing, model training, and evaluation.

RESEARCH PROJECTS

Reasoning-Based Image Geolocation with Vision-Language Models

- Studied whether VLMs can geolocate images by **reasoning over scene clues** rather than by direct classification; implemented a **GeoReasoner**-style pipeline.
- **Fine-tuned Qwen2-VL** on the **NaviClues** expert-reasoning dataset; improved accuracy further by adding **retrieval** with a **fine-tuned CLIP** encoder.

Vision-Based Retrieval-Augmented Generation (VisRAG)

- Implemented an end-to-end pipeline that retrieves document *images* rather than OCR text, using a **Qwen2-VL**-based retriever (**VisRAG-Ret**) and **MiniCPM-V 2.0** for generation.
- Benchmarked against an OCR-based **TextRAG** baseline on image document question answering to measure information lost through text extraction.

Improving Multi-View Consistency in Instruct-NeRF2NeRF

- Re-implemented Instruct-NeRF2NeRF for text-guided **3D scene editing** with **NeRFs** and **Instruct-Pix2Pix**.
- Found that the 2D edits are **inconsistent across views**; reduced this by modifying the iterative dataset-update scheme in the InstructPix2Pix editing loop.

Pretrained Hybrid Transformer–Mamba Language Models

- Re-implemented Manticore, which combines pretrained **Transformer** and **Mamba** blocks into hybrid models, then fine-tuned and evaluated the resulting hybrids.
- Trained the **Jamba** hybrid LLM across multiple GPUs with **DeepSpeed** pipeline parallelism.

Coding Agent Harness

- Built an **agent harness** that turns a single user prompt into a working end-to-end system, using **Gemini** tool specifications and wrappers for interchangeable LLM backends.

TEACHING

University of Wisconsin–Madison — Graduate Teaching Fellow *Aug 2023 – May 2026*

- **CS300**: Programming II (OOP & Data Structures), class of over **800** students. **Fall 2023, Spring 2024, Fall 2024, Spring 2025**
- **CS200**: Programming I (Introduction to Object-Oriented Programming). **Fall 2025**
- **CS600**: Software Engineering Industry Capstone Project. **Spring 2026**

INDUSTRY EXPERIENCE

Amazon AWS - Seattle, Washington *June – August 2026*
Software Engineering

- Built **agentic AI** internal tools for automation; implemented data structures for network-layer abstraction and deployed AWS network infrastructure to new regions.

Amazon - Seattle, Washington *May 2025 – August 2025*
Software Engineering Intern

- Upgraded the **AlexaExternalMedia Player** module to the latest SDK for recent Android and FireTV OS releases; built a cross-platform **Digital WellBeing** app that monitors streaming-app usage.

A2SV (Africa to Silicon Valley) – Palo Alto, CA *Dec 2021 – March 2023*
Software Engineer

- **Lead back-end developer** on HakimHub (healthcare platform) and internal tools in Node.js, NestJS, and PostgreSQL; deployed on Google Cloud, reducing latency by **20%** and post-deployment issues by **30%**.

Cybersoft PLC - Addis Ababa, Ethiopia *Sep 2020 – Mar 2021*
Software Engineering Intern

- Built a **full-stack** Land Management System; cut SQL retrieval times by **20%** through indexing and caching.

AWARDS

- Four-Year **Graduate Teaching Fellowship**, University of Wisconsin–Madison Aug 2023
- BSc in Software Engineering **with Distinction**, Addis Ababa University Oct 2021

MEMBERSHIP & SERVICE

- Member and Mentor at **Black in AI**, which increases the presence and inclusion of Black people in AI through idea-sharing, collaboration, mentorship, and advocacy.

SKILLS

- **ML / Research:** PyTorch, TensorFlow, DeepSpeed, Qwen2-VL, CLIP, NeRF, diffusion models, RAG, LLM fine-tuning, Pandas, NumPy
- **Programming Languages:** Python, C++, Java, Kotlin, JavaScript, SQL
- **Frameworks and Tools:** GCP, AWS, Android, NodeJS, Flask, MongoDB, ReactJS